

SURFACE CURRENTS

Surface currents are driven by the winds. They carry cold water to the tropics and warm water to the poles.

→ Warm current

→ Cold current

North Atlantic

Warm water is cooled by the ice of the Arctic and begins to sink.

Gyre

Surface currents, driven by winds and by the spin of planet Earth, often form circular patterns called gyres. Gyres north of the equator move clockwise, while those in the south move counterclockwise.

Ocean in motion

Ocean waters are constantly moving. Their movements, called currents, are driven by wind and the Earth's spin. But ocean currents are also affected by the water's temperature and saltiness, as well as sea depth.

OCEANIC CONVEYOR

Surface currents and deep ocean currents link up to form a planet-wide conveyor belt flowing at times across the ocean basins, then rising to the surface, before sinking again to the deep ocean floor.

